

AI Image Detector - Training Report (Vast.ai RTX 4090)

1. Executive Summary

This report documents the training of an EfficientNet-V2-S deep learning model for AI-generated image detection on a Vast.ai server equipped with an NVIDIA RTX 4090 GPU (24GB VRAM). The model was trained on the CIFAKE dataset (100,000 training images, 20,000 validation images) for 30 epochs, achieving a best validation accuracy of 98.97% and ROC AUC of 0.9974.

2. Hardware & Environment

GPU: NVIDIA GeForce RTX 4090 (24 GB VRAM)

CUDA Version: 12.7

Driver Version: 565.57.01

PyTorch: 2.x with CUDA + AMP (mixed precision)

Platform: Linux (Vast.ai)

Total Training Time: 80.7 minutes

3. Training Configuration

Model: EfficientNet-V2-S (20.8M parameters)

Dataset: CIFAKE (dragonintelligence/CIFAKE-image-dataset)

Train Samples: 100,000

Val Samples: 20,000

Image Size: 224 x 224

Batch Size: 64

Gradient Accumulation: 1 (effective batch = 64)

Optimizer: AdamW (weight_decay=0.01)

Learning Rate: 3e-4 (cosine schedule with 3-epoch warmup)

Loss Function: Label Smoothing Cross-Entropy (smoothing=0.1)

Data Augmentation: Strong (RandomResizedCrop, ColorJitter, GaussianBlur, RandomErasing)

Mixed Precision (AMP): Enabled

Early Stopping Patience: 7 epochs

Epochs Completed: 30 / 30 (no early stopping triggered)

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4. Final Results

Best Validation Accuracy: 98.97% (Epoch 29)

Best Validation Loss: 0.338921

Final Train Accuracy: 99.91%

Final Val Accuracy: 98.91%

ROC AUC Score: 0.9974

5. Classification Report (on 20K validation set)

	precision	recall	f1-score	support
Real	0.99	0.99	0.99	10000
AI-Generated	0.99	0.99	0.99	10000
accuracy			0.99	20000
macro avg	0.99	0.99	0.99	20000
weighted avg	0.99	0.99	0.99	20000

6. Epoch-by-Epoch Summary

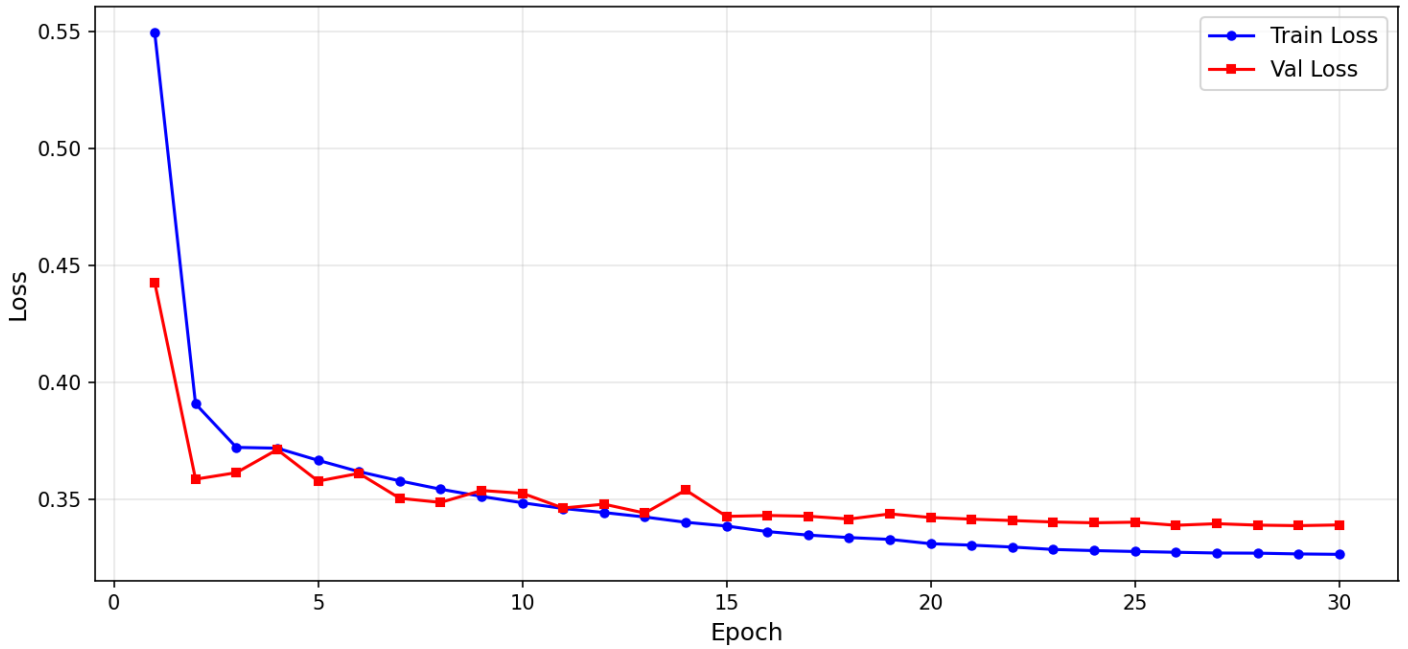
Ep	Train Loss	Train Acc	Val Loss	Val Acc	LR	Best
1	0.549705	79.73%	0.442875	90.07%	1.02e-04	*
2	0.390948	94.48%	0.358778	97.28%	2.01e-04	*
3	0.372356	96.09%	0.361558	96.90%	3.00e-04	
4	0.372011	96.20%	0.371403	95.97%	2.99e-04	
5	0.366837	96.60%	0.357964	97.32%	2.96e-04	*
6	0.362009	96.99%	0.361255	96.97%	2.91e-04	
7	0.358048	97.32%	0.350613	98.03%	2.84e-04	*
8	0.354491	97.60%	0.348844	97.98%	2.75e-04	
9	0.351412	97.91%	0.353938	97.61%	2.65e-04	
10	0.348701	98.12%	0.352755	97.56%	2.53e-04	
11	0.346263	98.32%	0.346445	98.19%	2.40e-04	*
12	0.344517	98.46%	0.348102	98.17%	2.25e-04	
13	0.342615	98.61%	0.344291	98.45%	2.10e-04	*
14	0.340373	98.77%	0.354121	97.65%	1.93e-04	
15	0.338762	98.95%	0.342879	98.58%	1.76e-04	*
16	0.336389	99.11%	0.343262	98.54%	1.59e-04	
17	0.334892	99.25%	0.342957	98.56%	1.42e-04	
18	0.333834	99.34%	0.341746	98.66%	1.25e-04	*
19	0.333049	99.40%	0.343939	98.57%	1.08e-04	
20	0.331210	99.55%	0.342399	98.67%	9.13e-05	*
21	0.330587	99.58%	0.341693	98.69%	7.57e-05	*
22	0.329780	99.66%	0.341102	98.78%	6.12e-05	*
23	0.328749	99.74%	0.340477	98.79%	4.79e-05	*
24	0.328263	99.79%	0.340140	98.88%	3.60e-05	*
25	0.327881	99.82%	0.340401	98.82%	2.56e-05	
26	0.327556	99.83%	0.339079	98.92%	1.69e-05	*
27	0.327236	99.86%	0.339758	98.92%	1.00e-05	
28	0.327173	99.87%	0.339144	98.92%	5.03e-06	
29	0.326841	99.90%	0.338921	98.97%	2.01e-06	*
30	0.326667	99.91%	0.339232	98.91%	1.00e-06	

* = new best validation accuracy

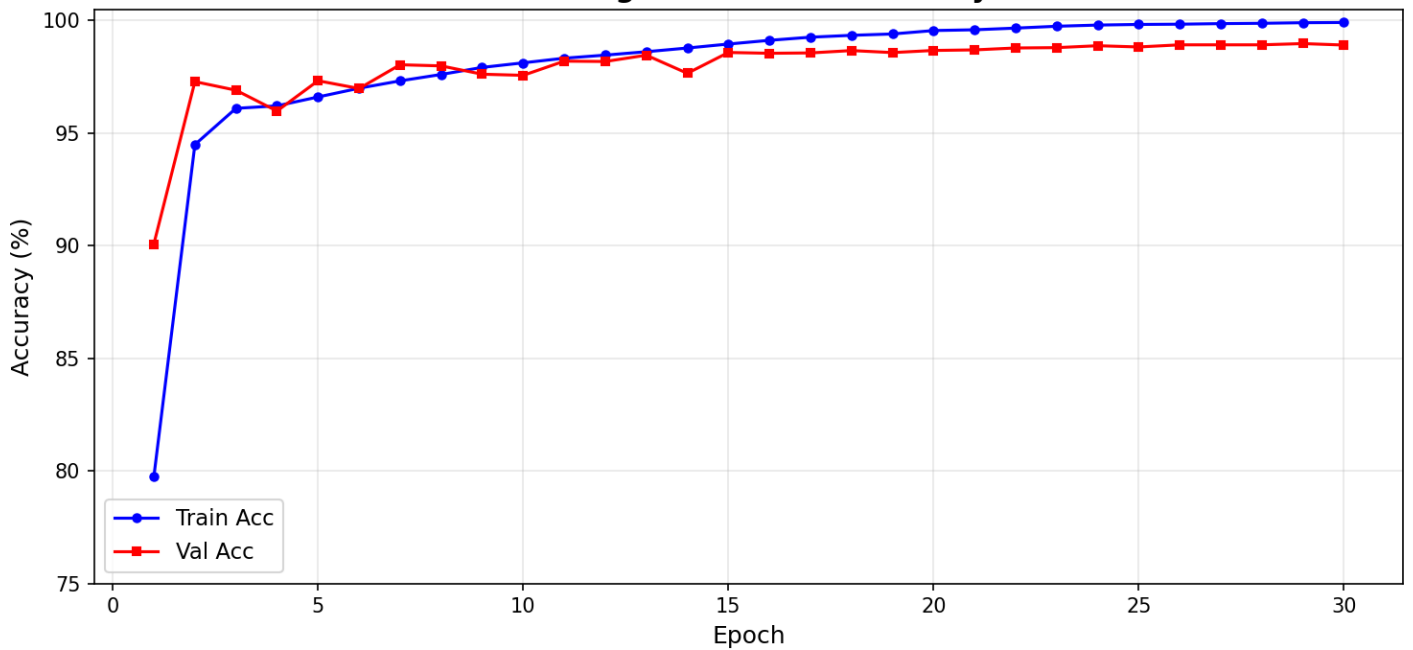
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7. Training Curves

Training & Validation Loss

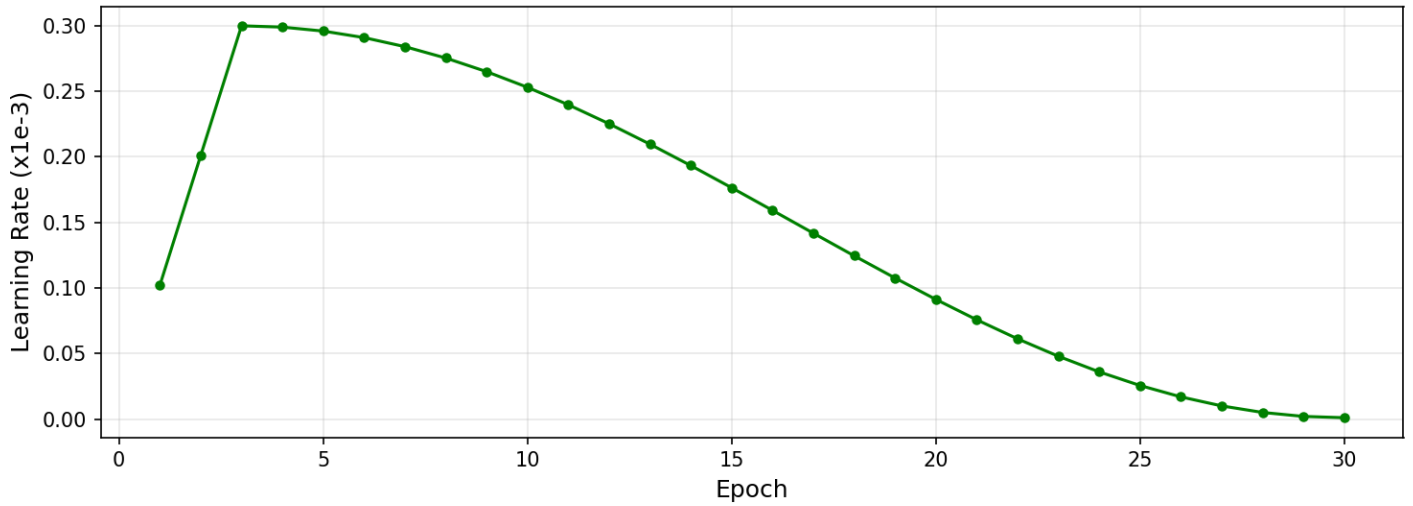


Training & Validation Accuracy

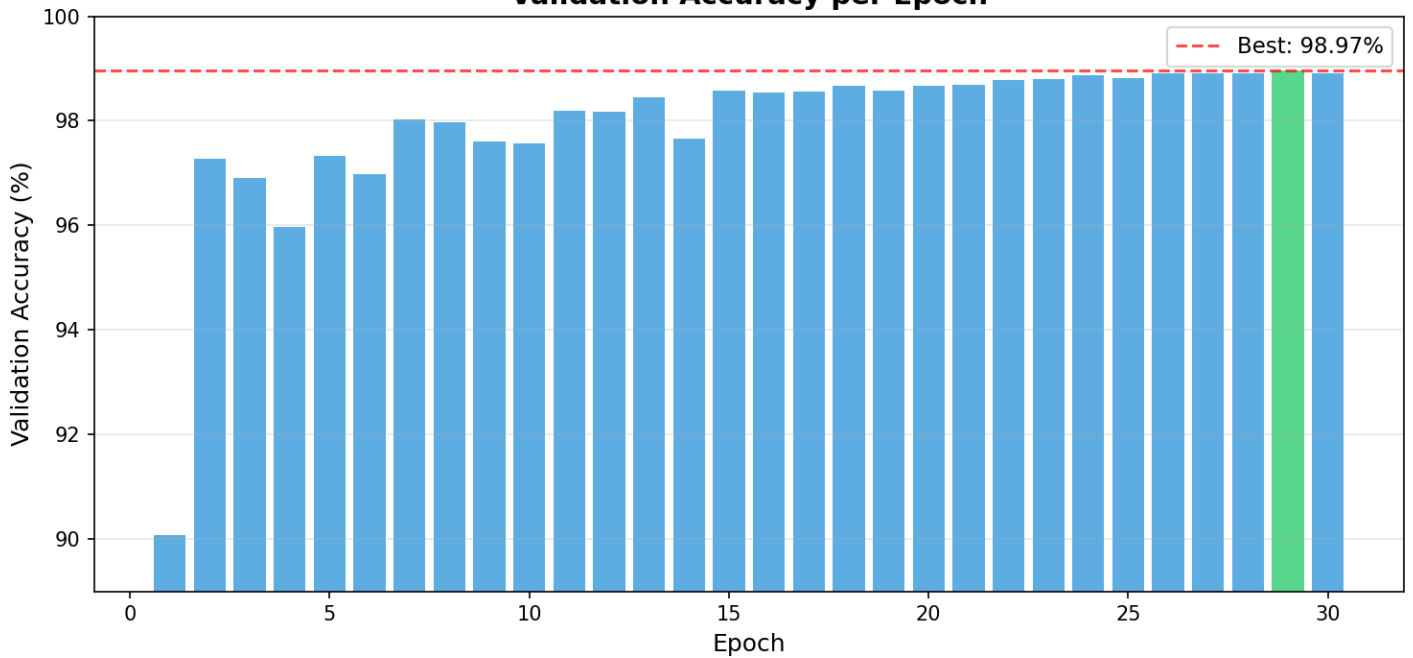


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Learning Rate Schedule (Cosine with Warmup)



Validation Accuracy per Epoch



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8. Errors Encountered

No errors were encountered during the training run. The entire pipeline (data loading from HuggingFace, model creation, training loop, evaluation, and checkpoint saving) executed successfully without any modifications required.

Minor notes:

- A HuggingFace warning about unauthenticated requests was displayed (non-blocking).
- numpy was downgraded from 2.4.3 to 1.26.4 during pip install to satisfy the requirements.txt constraint (numpy>=1.24.0,<2.0.0). This was handled automatically by pip.
- No CUDA out-of-memory issues with batch_size=64 on the RTX 4090.

9. Code Changes Made

No code changes were necessary. The repository's training pipeline ran cleanly out of the box on the Vast.ai RTX 4090 server. The following command was used:

```
python train_deep_cnn.py \  
  --model efficientnet --variant small \  
  --batch-size 64 --accumulation-steps 1 \  
  --epochs 30 --lr 3e-4 \  
  --num-workers 8 \  
  --loss label_smooth --label-smoothing 0.1 \  
  --early-stopping 7 \  
  --strong-augmentation \  
  --checkpoint-dir ./models/deep
```

Key differences from default settings:

- batch-size increased from 16 to 64 (RTX 4090 has 24GB VRAM)
- accumulation-steps reduced from 2 to 1 (not needed with large batch)
- num-workers increased from 2 to 8 (server has more CPU cores)

10. Key Observations

1. Rapid Convergence: The model reached 90% val accuracy after just 1 epoch and 97%+ after epoch 2, demonstrating the effectiveness of transfer learning from ImageNet-pretrained EfficientNet-V2-S.
2. Consistent Improvement: Validation accuracy improved steadily from 90.07% (epoch 1) to 98.97% (epoch 29), with the cosine LR schedule providing smooth convergence.
3. No Overfitting: The gap between train accuracy (99.91%) and val accuracy (98.97%) remained small (~1%), indicating good generalization aided by label smoothing and strong data augmentation.
4. Balanced Performance: The model achieves 0.99 precision and recall for BOTH classes (Real and AI-Generated), with no class bias.
5. Training Speed: The RTX 4090 completed all 30 epochs over 100K images in ~81 minutes, averaging ~2.7 minutes per epoch.